

Teacher:	Ray Joshua Balingkit	**Subject:**	Design & Technology
Week:	10 (Term 1)	**Year:**	10–11
Link:		**Cover:**	

LESSON OVERVIEW

What should students know/do by the end of this week's lessons?

Students will identify input, process and output in simple electronic systems. They will develop their understanding through a teacher explanation, a worked example and an evidence-based classroom task. In the second lesson, students will plan a simple control system with a clear decision rule and test cases. Their recorded reasoning, practical evidence and final checks will show whether they can apply the ideas to a defined user, product or design problem.

End-of-Week Assessment/Check: How will you know they got it?

Use teacher questioning, labelled diagrams, case comparisons and the completed independent tasks. Check accurate terminology, evidence linked to the user or brief and a reasoned explanation of choices. Review recorded tests or traces alongside conclusions, and use the final check in both lessons to identify the next skill needing practice.

LESSON SLOTS - TUESDAY AND WEDNESDAY

Lesson	Date	Learning Objective	The Lesson Flow	Differentiation
1	27 October 2026	By the end of this lesson, students will be able to identify input, process and output in simple electronic systems.	**H:** Present a diagram of an automatic night light. Ask how it can respond to darkness and why a battery alone cannot decide when the lamp should switch on. **ID:** Introduce a system as linked input, process and output functions, with a power supply providing energy. Explain sensors, a decision or processing stage and outputs such as light, sound or motion. Distinguish the information from a sensor from the energy needed by the output. **WD:** Annotate a block diagram for a light-controlled lamp. Label the light sensor as input, the threshold decision as process and the lamp as output. Model daytime and nighttime cases and explain that the physical circuit must be designed to implement the intended	**Support:** Provide component pictures, colour-coded blocks and an example rule. Complete one diagram together, then remove the labels for a different product. **Extension:** Students identify an example with two inputs and explain how combining them could alter the output, stating the decision rule clearly.

			rule. **YD:** Students classify components in three teacher-prepared product diagrams and draw an input-process-output model for one. They describe what happens when the input changes and identify the power source separately. A peer checks whether the process is an actual rule rather than simply the word computer. **C:** Ask students why the sensor is not the output and what the processor must decide. Check the block diagrams and use a contrasting temperature-controlled example to test transfer.	
2	28 October 2026	By the end of this lesson, students will be able to plan a simple control system with a clear decision rule and test cases.	**H:** Ask whether a classroom warning device should sound whenever any movement occurs. Use the responses to show why the intended context must be defined before choosing inputs. **ID:** Set a brief for a model desk-light controller that turns a light on when the room is dark. Explain system boundaries, threshold conditions and expected output states. Keep the design at block-diagram or simulation level; no mains wiring or unknown circuits are needed. **WD:** Model a sensor reading below a stated threshold and trace the resulting lamp state. Add test cases below, exactly at and above the threshold. Explain why the boundary case depends on the chosen comparison and must be specified rather than guessed. **YD:** Students plan an input-process-output	**Support:** Provide a partially completed block diagram and a table of sensor readings. Begin with two clear states before adding the exact threshold case. **Extension:** Students propose a second input or a delay to address an unwanted response, explaining how the added condition would change their test plan.



			system for the supplied brief, write its decision rule and produce at least three test cases with expected outputs. Partners trace the rule and identify an ambiguous condition. Students revise the rule and explain how the correction affects a test case. **C:** Check that the rule and expected outputs agree, particularly at the boundary. Ask what happens when the condition is false and identify plans that leave that state undefined.	
3				Support: Extension:
4				Support: Extension:

MATERIALS NEEDED FOR THE WEEK

- Pearson Edexcel GCSE (9–1) Design and Technology revision notes: 1.6 Electronic systems
- Teacher computer, display screen and whiteboard markers
- Prepared product scenarios, diagrams and user briefs aligned to the lesson objectives
- Prepared analysis worksheet, comparison grid or test-recording table
- Paper, pencils, rulers, card templates and approved classroom stationery
- Teacher-prepared samples or component cards; pre-cut model parts where required
- Lesson success criteria, observation checklist and labelled evidence folders



END-OF-WEEK REFLECTION

- What worked well?

- What flopped or confused students?

- What needs to be re-taught or adjusted next week?